
Annex 5 — Sector: Municipal Solid Waste (France)

Purpose: Sector-specific feedstock profile for French Municipal Solid Waste (residual household waste (OMR)), detailing composition and pyrolysis product yields when processed through the Urban Rig, with full justification and sourcing of every parameter.

Date: March 2026 **Status:** Corrected + Verified — Session 10 (full 13-category decomposition)

OBJECTIVE

To answer the question: **If we feed French MSW (residual household waste (OMR)) into the Urban Rig, what are the expected outputs?**

This annex is divided into two parts. **Part 1** establishes the composition of what actually enters the reactor — tracing from the full 13-category MODECOM characterisation through inert removal and material-by-material decomposition to a consolidated feedstock breakdown. **Part 2** then applies per-material pyrolysis yields (measured on the UR where available, literature-based elsewhere) to that feedstock to calculate weighted-average outputs.

Every parameter is classified as: **FACT** (measured on UR), **HYPOTHESIS** (literature-based estimate), or **ASSUMPTION** (engineering judgment).

PART 1 — WHAT GOES IN: Feedstock Composition

This part answers: of the 1,000 kg of raw MSW arriving at the gate, what materials actually enter the reactor, and in what proportions?

Why it matters: the financial model requires a precise breakdown by material type because each material produces different yields. Getting the input wrong cascades through every output calculation.

1. FRENCH OMR COMPOSITION — FULL 13-CATEGORY BREAKDOWN

Why we need this: The starting point for the entire model. MODECOM 2017 characterises French OMR into 13 categories. Previous versions of this annex used a simplified 9-category grouping that lumped five distinct categories into “Other/Fines” — losing visibility on ~190 kg of material per tonne. The full breakdown enables proper decomposition and eliminates the opaque residual.

Source: ADEME MODECOM 2017, France’s national waste characterisation campaign (third edition), characterising OMR after separate collection. Published at ademe.fr; detailed analysis at medias.amf.asso.fr.

#	Category	% of OMR	kg per 1,000 kg	Note
1	Putrescibles (food/garden waste)	27%	270	Largest single fraction
2	Plastics	15%	150	All polymer types mixed
3	Sanitary textiles	14%	140	Diapers, toilet paper, wipes, etc.
4	Fine elements (<20 mm)	10%	100	Sub-sieve fraction, mix of everything
5	Paper	8%	80	Newspapers, magazines, office paper
6	Cardboard	6%	60	Packaging cardboard
7	Glass	5%	50	Bottles, jars

#	Category	% of OMR	kg per 1,000 kg	Note
8	Textiles (clothing)	4%	40	Clothing, fabric scraps
9	Composites (EMR)	3%	30	Multi- material packag- ing (Tetra Pak, etc.)
10	Metals	3%	30	Ferrous + non- ferrous
11	Unclassified incombustibles	2%	20	Ceramics, stones, porcelain
12	Wood	1.5%	15	Wood frag- ments
13	Unclassified combustibles	1.5%	15	Leather, rubber, misc. organic
TOTAL		100%	1,000	

[FACT] **FACT** — These percentages come from a statistically representative national study. Data quality: HIGH. Categories 4–13 were previously grouped as “Other/Fines (19%)” and “Composites (3%)” — now shown individually.

Note on MODECOM 2024 update: The latest campaign (valor3e.fr, 2024 first results) shows further evolution: biowaste rising to 32%, sanitary textiles to 25.5%, recyclables at 24.1%. OMR per capita declined from 254 kg (2017) to 223.5 kg (2024). The underlying trends would not materially change the yield model since the plastic-like fraction ratios remain similar.

2. REMOVING NON-PROCESSABLE MATERIAL

Why we need this: Several categories cannot be pyrolysed and must be removed before or during processing. Quantifying all removals — not just glass and metals

— gives an accurate reactor input mass.

Removed material	Mass (kg)	Removal method
Glass (glass)	50	Mechanical screening / optical sorting
Metals (ferrous + non-ferrous)	30	Magnetic + eddy current separation
Unclassified incombustibles	20	Density separation / manual sorting
Aluminium from composites (5% of 30 kg)	1.5	Separated during composite processing
Metal fragments from fines (~3% of 100 kg)	3.0	Magnetic separation of fines
Glass fragments from fines (~5% of 100 kg)	5.0	Screening of fines
Total removed	109.5	
Material entering reactor	890.5	

[HYPOTHESIS] **ASSUMPTION** — Pre-sorting assumed at standard industry efficiency. The fines fragment estimates (3% metal, 5% glass) are based on published characterisation studies of the sub-20mm fraction. In practice, removal efficiency varies; residual fragments are captured in the mineral inert residue.

Note: Previous versions removed only glass (50 kg) and metals (30 kg) = 80 kg, giving 920 kg effective feedstock. The fuller accounting adds 29.5 kg of removals from incombustibles, composite aluminium, and fines fragments.

3. SANITARY TEXTILES DECOMPOSITION (140 kg)

Why we need this: MODECOM groups “sanitary textiles” as a single category (14% of OMR), but this category contains radically different materials — from pure cellulose (toilet paper) to synthetic polymers (diaper components). Since each material has different pyrolysis yields, we must decompose this category into its constituent materials.

3.1 Subcategory Breakdown

Source: vertuow.com/tsuu-transformer-les-dechets-en-ressources.

Subcategory	% of sanitary textiles	Mass in 140 kg	Dominant material
Toilet paper	44.3%	62.0 kg	100% cellulose
Paper towels	17.2%	24.1 kg	100% cellulose
Baby diapers	10.7%	15.0 kg	SAP + PP + PE + cellulose
Body wipes	5.4%	7.6 kg	PP/PET nonwoven + moisture
Adult incontinence products	~4%	~5.6 kg	SAP + PP + PE + cellulose
Other (protections, cotton buds, etc.)	~18.4%	~25.8 kg	Mixed cellulose, cotton, some synthetic

[FACT] **FACT** — Subcategory percentages from French waste-sector analysis. Data quality: MEDIUM-HIGH.

3.2 Diaper Material Composition

Sources: ResearchGate material composition data; ChemistryExplained.com.

Component	% of diaper mass	Type
Cellulose pulp (absorbent core)	37%	Organic
SAP — sodium polyacrylate	31%	Synthetic polymer
Polypropylene (top sheet, elastic)	16%	Plastic
LDPE (backsheet)	6%	Plastic
Tape, adhesive, elastic	10%	Mostly synthetic

3.3 Material Reassignment

Material	Mass (kg)	Reassigns to
Cellulose from toilet paper	62.0	→ Cellulose
Cellulose from paper towels	24.1	→ Cellulose

Material	Mass (kg)	Reassigns to
Cellulose from diapers (37% of 20.6 kg)	7.6	→ Cellulose
Cellulose/cotton from other items (~50%)	16.7	→ Cellulose
Total cellulose	110.4	79% of TS
SAP from diapers (31% of 20.6 kg)	6.4	→ Plastic-like
PP/PE from diapers (22% of 20.6 kg)	4.5	→ Plastic-like
PP/PET nonwoven from wipes (80% of 7.6 kg)	6.1	→ Plastic-like
Other synthetic from misc. (~15%)	5.0	→ Plastic-like
Total plastic-like	22.0	16% of TS
Moisture/contamination	7.6	→ Organic (non-productive)

[HYPOTHESIS] **ASSUMPTION** — The 50% cellulose and 15% synthetic estimates for “other” are engineering judgments.

4. TEXTILES — CLOTHING DECOMPOSITION (40 kg)

Why we need this: Clothing textiles are 4% of OMR — entirely absent from previous versions of this model, where they were hidden inside “Other/Fines.” They contain significant quantities of both cellulose (cotton) and synthetic polymers (polyester, nylon) that must be assigned to the correct material group.

European textile waste composition (source: EEA Textiles and the Environment report, 2024; cross-checked with Euratex industry data):

Fibre type	% of textile waste	Mass in 40 kg	Reassigns to
Cotton	~60%	24.0 kg	→ Cellulose
Polyester (PET-like)	~21%	8.4 kg	→ Plastic-like
Nylon (polyamide)	~9%	3.5 kg	→ Plastic-like
Acrylic / other synthetic	~5%	2.1 kg	→ Plastic-like
Non-textile (buttons, zips, etc.)	~5%	2.0 kg	→ Mineral inert

[HYPOTHESIS] **ASSUMPTION** — European average fibre mix applied to French OMR textile fraction. Actual proportions vary by collection area and season.

5. COMPOSITES DECOMPOSITION (30 kg)

Why we need this: MODECOM “composites” are overwhelmingly multi-material packaging — Tetra Pak-type beverage cartons, blister packs, etc. Their composition is well-documented and can be decomposed with high confidence.

Tetra Pak composition (source: Tetra Pak sustainability reports; industry standard):

Component	% of composite mass	Mass in 30 kg	Reassigns to
Cardboard	~75%	22.5 kg	→ Cellulose
Polyethylene (PE film)	~20%	6.0 kg	→ Plastic-like (LDPE)
Aluminium foil	~5%	1.5 kg	→ Removed (non-processable)

[FACT] **FACT** — Tetra Pak composition is an industry standard. Data quality: HIGH.

6. UNCLASSIFIED COMBUSTIBLES DECOMPOSITION (15 kg)

Why we need this: This catch-all category (leather, rubber, wood offcuts, misc. organic material) was previously invisible inside “Other/Fines.” While small, it contains both organic and synthetic fractions that behave differently in the reactor.

Component	% estimate	Mass in 15 kg	Reassigns to
Leather, rubber (organic/protein)	~70%	10.5 kg	→ Organic
Synthetic (rubber, adhesives, misc.)	~30%	4.5 kg	→ Plastic-like

[HYPOTHESIS] **ASSUMPTION** — Split based on general waste characterisation literature. Low confidence due to heterogeneity.

7. FINE ELEMENTS DECOMPOSITION (100 kg → 92 kg after fragment removal)

Why we need this: The sub-20mm sieve fraction is the second-largest “hidden” category after refus. At 10% of OMR, it contains meaningful quantities of every material type. Without decomposing it, 100 kg per tonne has undefined pyrolysis behaviour. After removing metal fragments (3 kg) and glass fragments (5 kg) in Section 2, 92 kg enters the reactor.

Composition of the sub-20mm fraction (source: published MODECOM characterisation studies of fine elements; cross-validated with local characterisation data from French waste operators):

Component	% of fines	Mass in 92 kg	Reassigns to
Organic fragments (food scraps)	~45%	41.4 kg	→ Organic
Paper/cardboard scraps	~15%	13.8 kg	→ Cellulose
Small plastic fragments	~8%	7.4 kg	→ Plastic-like (MSW polymer mix)
Mineral inert (soil, dust, ceramics)	~32%	29.4 kg	→ Mineral inert (enters reactor, produces only residue)

[HYPOTHESIS] **ASSUMPTION** — Percentages based on published characterisation data. The mineral inert fraction enters the reactor but passes through as residue without producing oil or useful gas. MEDIUM confidence.

8. PLASTIC POLYMER MIX — FULL PLASTIC-LIKE FRACTION

Why we need this: The plastic-like fraction now totals 203.9 kg (was 176.2 kg in previous version). The additional 27.7 kg comes from textiles, composites, combustibles, and fines. Each source has a different polymer profile, which affects the weighted oil yield. This section consolidates all plastic-like material by polymer type.

8.1 European Plastic Packaging Waste Composition

Source: PlasticsEurope 2017, via [pmc.ncbi.nlm.nih.gov/articles/PMC8162419](https://pubmed.ncbi.nlm.nih.gov/articles/PMC8162419).

Polymer	% of plastic waste
LDPE	32%
HDPE	20%
PP	20%
PET	18%
PS	7%
PVC	3%

8.2 Full Plastic-Like Fraction by Polymer (203.9 kg)

Polymer	From MODE- COM plastics (150 kg)	From TS (22 kg)	From textiles (14 kg)	From com- posites (6 kg)	From com- bustibles (4.5 kg)	From fines (7.4 kg)	TOTAL
LDPE	48.0	4.2	—	6.0	—	2.4	60.6
HDPE	30.0	—	—	—	—	1.5	31.5
PP	30.0	7.5	—	—	—	1.5	39.0
PET	27.0	3.1	—	—	—	1.3	31.4
PS	10.5	—	—	—	—	0.5	11.0
PVC	4.5	—	—	—	—	0.2	4.7
SAP	—	6.4	—	—	—	—	6.4
Polyester	—	—	8.4	—	—	—	8.4
Nylon (PA)	—	—	3.5	—	—	—	3.5
Other synthetic	—	0.8*	2.1	—	4.5	—	7.4
TOTAL	150.0	22.0	14.0	6.0	4.5	7.4	203.9

*TS “other synthetic” (5.0 kg) split: 4.2 kg LDPE (diaper backsheet) already counted in LDPE row; 0.8 kg remaining misc.

Key takeaway: The polyolefin fraction (LDPE + HDPE + PP) = 131.1 kg, or 64% of the plastic-like total. These are the highest-yield polymers (77–80% oil). The remaining 36% includes lower-yield materials (PET, PS, PVC, nylon, SAP, misc.) that pull the weighted average down.

9. CONSOLIDATED EFFECTIVE FEEDSTOCK — FINAL TABLE

Why we need this: This is the final output of Part 1 — the complete material-by-material breakdown of what enters the reactor. All 1,000 kg of raw OMR is now accounted for: 109.5 kg removed, 890.5 kg enters the reactor, of which 859.1 kg is processable material and 31.4 kg is mineral inert that passes through as residue.

9.1 Four-Category View (890.5 kg entering reactor)

Category	Mass (kg)	% of reactor input	Composition
Organic	329.5	37.0%	Putrescibles 270 + fines organic 41.4 + combustibles organic 10.5 + TS moisture 7.6
Cellulose	325.7	36.6%	Paper 80 + cardboard 60 + TS cellulose 110.4 + textile cotton 24 + composite cardboard 22.5 + wood 15 + fines paper 13.8
Plastic-like	203.9	22.9%	See polymer-by- polymer table above (Section 8.2)
Mineral inert	31.4	3.5%	Fines mineral 29.4 + textile non-textile 2.0
TOTAL	890.5	100%	

9.2 Three-Category View (859.1 kg processable material)

This is the view used in Annex 2 — showing only the three material groups that produce useful pyrolysis outputs, excluding mineral inert.

Category	Mass (kg)	% of processable feedstock	Primary Urban Rig Output
Organic (food, garden, misc.)	329.5	38.4%	Charcoal / biochar
Cellulose (paper, card, wood, cotton)	325.7	37.9%	Charcoal + off-gas
Plastic- like (all poly- mers)	203.9	23.7%	Fuel oil (4 fractions)
TOTAL process- able	859.1	100%	

9.3 Full Mass Balance

From → To	Mass (kg)
Raw OMR at gate	1,000.0
– Glass, metals, incombustibles, hazardous, fragments	–109.5
= Material entering reactor	890.5
– Mineral inert in reactor (produces only residue)	–31.4
= Processable feedstock	859.1

9.4 Traceability — Where Every Kilogram Comes From

Annex 5 category	Organic	Cellulose	Plastic-like	Mineral inert	Removed	Total
Putrescibles (270)	270.0	—	—	—	—	270
Paper (80)	—	80.0	—	—	—	80
Cardboard (60)	—	60.0	—	—	—	60
Sanitary textiles (140)	7.6	110.4	22.0	—	—	140
Plastics (150)	—	—	150.0	—	—	150
Textiles (clothing) (40)	—	24.0	14.0	2.0	—	40
Composites (30)	—	22.5	6.0	—	1.5	30
Wood (15)	—	15.0	—	—	—	15
Unclassified combustibles (15)	10.5	—	4.5	—	—	15
Glass (50)	—	—	—	—	50.0	50
Metals (30)	—	—	—	—	30.0	30
Incombustibles + hazardous (20)	—	—	—	—	20.0	20
Fine elements (100)	41.4	13.8	7.4	29.4	8.0	100
TOTAL	329.5	325.7	203.9	31.4	109.5	1,000

PART 2 — WHAT COMES OUT: Pyrolysis Yields

This part answers: for each material identified in Part 1, what does the Urban Rig produce? Then, what are the weighted-average outputs for the full feedstock?

Why it matters: the financial model's revenue projections depend on oil output (sold at ICIS PPO index prices), the off-gas energy balance determines fuel costs, and the residue/char fraction determines disposal costs and biochar revenue.

10. PER-MATERIAL PYROLYSIS YIELDS

Data source for UR-measured yields: Asada Shokai test programme (Noda, 2022) — 74-page test report with independent measurement by Land Brain Co., Ltd. Original document: Asada Shokai data 2.pdf.

Per-material yield parameters are documented in detail in Annex 2 §2.3. The values below are the same parameters applied to this feedstock.

10.1 Polyethylene (PE: LDPE + HDPE)

Parameter	Value	Classification	Source
Oil yield	77%	[FACT] FACT	Asada Shokai test #2 (PE molded, 40 kg → 30.7 kg oil)
Off-gas yield	22%	[FACT] FACT	Asada Shokai test #2 (40 kg → 8.8 kg off-gas)
Residue	1%	[FACT] FACT	Asada Shokai test #2 (40 kg → 0.5 kg residue)
Oil density	0.83 kg/L	[FACT] FACT	Asada Shokai measurement
Off-gas CV	5.54 MJ/kg	[FACT] FACT	Asada Shokai energy balance tables, pp. 35–36

Justification: PE is the best-performing polymer in the UR. Two tests (pellets and molded) gave 74–77% oil yield. The molded test at larger scale (40 kg) is the more reliable data point.

10.2 Polypropylene (PP)

Parameter	Value	Classification	Source
Oil yield	80%	[FACT] FACT	Asada Shokai test #6 (PP molded, 50 kg → 39.8 kg oil)
Off-gas yield	18%	[FACT] FACT	Asada Shokai test #6 (50 kg → 10.0 kg off-gas)
Residue	2%	[FACT] FACT	Asada Shokai test #6 (50 kg → 0.2 kg residue)

Justification: PP is the highest-yield polymer tested. PP and PE are both polyolefins with similar cracking behaviour.

10.3 Polystyrene (PS)

Parameter	Value	Classification	Source
Oil yield	58%	[FACT] FACT	Average of Asada Shokai tests #3 (61%) and #4 (55%)
Off-gas yield	14%	[FACT] FACT	Average of tests #3 and #4
Residue	28%	[FACT] FACT	Average of tests #3 and #4

Justification: PS performs significantly worse than polyolefins due to its aromatic ring structure. The applied test (PS foam/EPS, test #10) gave 73% oil, but for conservative modelling we use 58%.

10.4 Polyethylene Terephthalate (PET)

Parameter	Value	Classification	Source
Oil yield	45%	[HYPOTHESIS] HYPOTHESIS	Slow pyrolysis literature
Off-gas yield	17%	[HYPOTHESIS] HYPOTHESIS	Derived from mass balance
Residue/Char	38%	[HYPOTHESIS] HYPOTHESIS	Slow pyrolysis literature

[!] **PET was NOT tested by Asada Shokai on the UR equipment.**

Derivation: Primary source: *Polymer Degradation and Stability*, 2022 (sciencedirect.com/S0141391022000866) — slow pyrolysis of PET at 400°C gave 46.7% oil, 39.7% solid, ~13.3% gas. We use 45% (small discount for secondary cracking at UR's 600°C final temperature). Cross-check: PMC 7183261.

10.5 Polyvinyl Chloride (PVC)

Parameter	Value	Classification	Source
Oil yield	32%	[HYPOTHESIS] HYPOTHESIS	Two-stage pyrolysis literature
HCl (filtered)	53%	[FACT] FACT (chemistry)	Stoichiometric
Off-gas (non-HCl)	~2%	[HYPOTHESIS] HYPOTHESIS	Mass balance
Residue/Char	13%	[HYPOTHESIS] HYPOTHESIS	Two-stage literature

[!] **PVC was NOT tested by Asada Shokai on the UR equipment.** The UR includes a patented catalytic chlorine adsorption layer (UR Recycling Solutions ENG, pp. 14–15).

Derivation: Two-stage dechlorination (200–340°C) removes 53% of mass as HCl; remaining polyene backbone cracks to ~68% oil, ~28% char, ~4% gas. Net: $47\% \times 68\% = \sim 32\%$ oil. Sources: sciencedirect.com/S0048969723069723; PMC 12096105.

10.6 Polyester (from textiles)

Parameter	Value	Classification	Source
Oil yield	45%	[HYPOTHESIS] HYPOTHESIS	Same as PET (polyester = PET-family polymer)
Off-gas yield	17%	[HYPOTHESIS] HYPOTHESIS	Mass balance
Residue/Char	38%	[HYPOTHESIS] HYPOTHESIS	

Justification: Clothing polyester is polyethylene terephthalate — chemically identical to PET bottle resin. Same pyrolysis behaviour. Contamination from dyes and finishes may slightly increase char formation, but this is within the uncertainty range.

10.7 Nylon — Polyamide (from textiles)

Parameter	Value	Classification	Source
Oil yield	45%	[HYPOTHESIS] HYPOTHESIS	PA6/PA66 pyrolysis literature
Off-gas yield	20%	[HYPOTHESIS] HYPOTHESIS	Literature
Residue/Char	35%	[HYPOTHESIS] HYPOTHESIS	Literature

[!] **Nylon was NOT tested by Asada Shokai on the UR equipment.**

Justification: Nylon 6 pyrolysis at 500–700°C produces 40–80% caprolactam (monomer) depending on conditions, plus secondary products. For slow pyrolysis at UR conditions (~600°C, steam carrier), a conservative 45% total oil estimate accounts for both monomer recovery and secondary liquid products. Similar to PET as both are condensation polymers with heteroatoms (N for nylon, O for PET).

10.8 SAP — Sodium Polyacrylate (from diapers)

Parameter	Value	Classification	Source
Oil yield	20%	[HYPOTHESIS] HYPOTHESIS	General polymer decomposition literature
Off-gas yield	30%	[HYPOTHESIS] HYPOTHESIS	Estimated
Residue	50%	[HYPOTHESIS] HYPOTHESIS	Na ₂ CO ₃ inorganic ash + char

Justification: Crosslinked acrylic polymer. Sodium content means ~25-30% ends up as solid inorganic ash regardless of temperature.

10.9 Other Synthetic Materials

Parameter	Value	Classification	Source
Oil yield	25%	[HYPOTHESIS] HYPOTHESIS	Blended estimate for mixed adhesives, elastics, rubber
Off-gas yield	25%	[HYPOTHESIS] HYPOTHESIS	
Residue	50%	[HYPOTHESIS] HYPOTHESIS	

10.10 Cellulose (Paper, Cardboard, Wood, Cotton)

Parameter	Value	Classification	Source
Oil yield	12%	[HYPOTHESIS] HYPOTHESIS	Biomass pyrolysis literature
Off-gas yield	20%	[HYPOTHESIS] HYPOTHESIS	Literature range 15–25%

Parameter	Value	Classification	Source
Residue/Char	68%	[HYPOTHESIS] HYPOTHESIS	Cellulose char formation at 600°C

Justification: Cellulose pyrolysis produces mainly char and gas. Bio-oil has low calorific value (~17–20 MJ/kg vs ~43 MJ/kg for plastic oil). Sources: Bridgwater, *Biomass and Bioenergy*, 2012; Chen et al., *Waste Management*, 2014.

10.11 Putrescibles (Food/Garden Waste)

Parameter	Value	Classification	Source
Oil yield	10%	[HYPOTHESIS] HYPOTHESIS	Biomass pyrolysis literature
Off-gas yield	20%	[HYPOTHESIS] HYPOTHESIS	
Residue/Char	70%	[HYPOTHESIS] HYPOTHESIS	High moisture → energy penalty

10.12 Mineral Inert (soil, dust, ceramics from fines)

Parameter	Value	Classification	Source
Oil yield	0%	[FACT] FACT	Inorganic — cannot pyrolyse
Off-gas yield	0%	[FACT] FACT	
Residue	100%	[FACT] FACT	Passes through unchanged

11. WEIGHTED YIELD CALCULATION

11.1 Plastic-Like Fraction (203.9 kg)

Material	Mass (kg)	Oil yield	Oil (kg)	Off-gas (kg)	Residue (kg)	Data quality
PE (LDPE+HDPE)	92.1	77%	70.9	20.3	0.9	[FACT] FACT
PP	39.0	80%	31.2	7.0	0.8	[FACT] FACT
PET	31.4	45%	14.1	5.3	11.9	[HYPOTHESIS] HYPOTHE- SIS
PS	11.0	58%	6.4	1.5	3.1	[FACT] FACT
PVC	4.7	32%	1.5	0.1†	0.6	[HYPOTHESIS] HYPOTHE- SIS
PVC → HCl	—	—	—	(2.5 filtered)	—	[FACT] FACT (stoich.)
Polyester	8.4	45%	3.8	1.4	3.2	[HYPOTHESIS] HYPOTHE- SIS
Nylon (PA)	3.5	45%	1.6	0.7	1.2	[HYPOTHESIS] HYPOTHE- SIS
SAP	6.4	20%	1.3	1.9	3.2	[HYPOTHESIS] HYPOTHE- SIS
Other synth.	7.4	25%	1.9	1.9	3.7	[HYPOTHESIS] HYPOTHE- SIS
SUBTOTAL	203.9		132.7	40.1	28.6	
<i>(PVC HCl filtered)</i>				<i>(2.5)</i>		
Mass check	203.9		132.7 + 40.1 + 28.6 + 2.5 = 203.9 ✓			

Material	Mass (kg)	Oil yield	Oil (kg)	Off-gas (kg)	Residue (kg)	Data quality
Weighted oil yield		65.1%	(132.7 / 203.9)			

† PVC off-gas excludes 2.5 kg HCl captured by the UR patented chlorine filtration system.

11.2 Organic + Cellulose Fraction (655.2 kg)

Material	Mass (kg)	Oil yield	Oil (kg)	Off-gas (kg)	Residue (kg)	Data quality
Cellulose	325.7	12%	39.1	65.1	221.5	[HYPOTHESIS] HYPOTHE-SIS
Putrescible organic	329.5	10%	33.0	65.9	230.7	[HYPOTHESIS] HYPOTHE-SIS
SUBTOTAL	655.2		72.0	131.0	452.2	

11.3 Mineral Inert Fraction (31.4 kg)

Material	Mass (kg)	Oil yield	Oil (kg)	Off-gas (kg)	Residue (kg)
Mineral inert	31.4	0%	0	0	31.4

11.4 TOTAL OUTPUT — 890.5 kg Entering Reactor

Product	From plastic-like	From or- ganic/cellulose	From mineral inert	TOTAL	% of 890.5 kg
Oil	132.7	72.0	0	204.7	23.0%
Off-gas	40.1	131.0	0	171.1	19.2%
Residue/Char	286	452.2	31.4	512.2	57.5%

Product	From plastic-like	From organic/cellulose	From mineral inert	TOTAL	% of 890.5 kg
HCl (filtered)	2.5	0	0	2.5	0.3%

Comparison with previous version (920 kg base):

Metric	Previous	Updated	Change
Total oil	184.8 kg (20.1%)	204.7 kg (23.0%)	+19.9 kg (+2.9 pp)
Total off-gas	175.2 kg (19.0%)	171.1 kg (19.2%)	−4.1 kg (+0.2 pp)
Total residue	557.5 kg (60.6%)	512.2 kg (57.5%)	−45.3 kg (−3.1 pp)
Plastic-like oil yield	66%	65.1%	−0.9 pp

The overall oil yield increases from 20.1% to 23.0% because material previously classified as unproductive “fines/other” is now properly allocated — much of it to cellulose and organic fractions that produce some oil, and some to plastic that produces significant oil.

12. OFF-GAS ENERGY

Off-gas energy modelling is consolidated in Annex 2 §2.5. The key parameter used in this feedstock model is 5.54 MJ/kg off-gas CV (constant across all feedstocks, Asada Shokai energy balance tables pp. 35–36).

13. CHAR/RESIDUE CHARACTERISATION AND ENERGY POTENTIAL

Char characterisation, calorific value estimates, commercialisation scenarios, and market valuation are consolidated in Annex 2 §2.4.3 and Annex 3 §3.2. The key finding: 480.8 kg combustible char per 1,000 kg OMR at weighted 15.6 MJ/kg CV

demonstrates fuel-grade quality suitable for commercial energy markets — see Annex 2 §2.6.3 for the full energy balance.

14. OIL YIELD SCENARIO COMPARISON

14.1 Three Operating Scenarios

Metric	Scenario A: Full OMR (with decomposition)	Scenario B: Plastic-like only	Scenario C: Polyolefins only (PE+PP)
Feed mass (per 1,000 kg OMR)	890.5 kg	203.9 kg	131.1 kg
Oil output	204.7 kg (23%)	132.7 kg (65%)	102.1 kg (78%)
Off-gas output	171.1 kg	40.1 kg	24.7 kg
Residue	512.2 kg (58%)	28.6 kg (14%)	2.4 kg (2%)
Off-gas energy	~484 MJ	~222 MJ	~137 MJ
Oil quality	Mixed (plastic + bio-oil)	Mostly hydrocarbon	High-quality hydrocarbon

14.2 Key Insights

1. **Full OMR gives 23% oil yield** (was 20.1% in previous version) because material previously hidden in “fines/other” is now properly allocated and contributes to oil production.

2. **Pre-sorting to isolate the plastic-like fraction raises oil yield to 65%.** Slightly lower than previous 66% because the expanded plastic-like fraction now includes lower-yield materials (polyester, nylon, misc. synthetics from textiles and combustibles).
3. **Polyolefin-only feed (PE+PP) gives 78% oil yield** — unchanged, consistent with the Asada Shokai mixed-polymer test (77%).
4. **The CDG airport context favours Scenario B or C.** Airport packaging waste is plastic-rich and can be pre-sorted.

14.3 Sensitivity: Impact of PET Yield Assumption

Since PET (+ polyester from textiles) was not tested on the UR and represents 19.5% of the plastic-like fraction:

PET oil yield assumption	Total plastic-like oil yield
30% (pessimistic)	62%
40% (moderate)	64%
45% (our estimate)	65%
55% (optimistic)	67%
70% (catalytic, upper lit.)	71%

15. CROSS-REFERENCE: ASADA SHOKAI MUNICIPAL WASTE TESTS

The Asada Shokai test programme included two tests with actual Japanese municipal waste (Noda city):

Test	Material	Oil %	Off-gas %	Foreign matter %	Char %
#14	Container recycling plastic	32%	19%	21%	28%
#15	Non-combustible residue	45%	22%	11%	23%

These results are **lower** than our Scenario B estimate (65%) because of foreign matter (11–21%), different plastic mix, and contamination. Adjusting for foreign matter: container recycling on plastic-only basis $\approx$ 63% — consistent with our estimate.

16. DATA QUALITY SUMMARY

Parameter	Classification	Confidence
French OMR composition (13 categories)	[FACT] FACT	HIGH — ADEME national study
Sanitary textiles subcategories	[FACT] FACT	MEDIUM-HIGH
Diaper material composition	[FACT] FACT	HIGH — peer-reviewed
European plastic polymer mix	[FACT] FACT	HIGH — PlasticsEurope
Textile fibre composition	[HYPOTHESIS] HYPOTHESIS	MEDIUM — European average
Composite (Tetra Pak) composition	[FACT] FACT	HIGH — industry standard
Fine elements composition	[HYPOTHESIS] ASSUMPTION	MEDIUM — characterisation studies
PE oil yield (77%)	[FACT] FACT	HIGH — measured on UR
PP oil yield (80%)	[FACT] FACT	HIGH — measured on UR
PS oil yield (58%)	[FACT] FACT	MEDIUM — range 55-73% across UR tests
Off-gas CV (5.54 MJ/kg)	[FACT] FACT	HIGH — consistent across 4 UR tests
PET oil yield (45%)	[HYPOTHESIS] HYPOTHESIS	MEDIUM — slow pyrolysis lit.
PVC oil yield (32%)	[HYPOTHESIS] HYPOTHESIS	MEDIUM — two-stage lit.
Polyester oil yield (45%)	[HYPOTHESIS] HYPOTHESIS	MEDIUM — same as PET
Nylon oil yield (45%)	[HYPOTHESIS] HYPOTHESIS	LOW-MEDIUM — PA literature
SAP oil yield (20%)	[HYPOTHESIS] HYPOTHESIS	LOW
Cellulose/putrescible yields	[HYPOTHESIS] HYPOTHESIS	MEDIUM — biomass lit.

Parameter	Classification	Confidence
Char CV — organic (15 MJ/kg)	[HYPOTHESIS] HYPOTHESIS	MEDIUM — MSW biomass char literature
Char CV — plastic (25 MJ/kg)	[HYPOTHESIS] HYPOTHESIS	MEDIUM — carbon black literature
Char energy self-sufficiency	[HYPOTHESIS] HYPOTHESIS	HIGH — robust under pessimistic assumptions

17. RECOMMENDATIONS FOR THE BUSINESS PLAN

1. **Use Scenario B (pre-sorted plastic-like fraction) as the base case** for oil yield projections: **65% oil yield on plastic-like feedstock, or ~23% on full OMR.**
2. **The 77-80% oil yield remains valid for clean polyolefin feed** (Scenario C).
3. **The three-category split for Annex 2** should use the fully derived percentages: **38.4% organic, 37.9% cellulose, 23.7% plastic** (on processable feedstock basis).
4. **Flag PET/polyester and nylon yields as needing UR-specific validation.** These now represent ~21% of the plastic-like fraction (was 17% before textiles were decomposed).
5. **The CDG feedstock advantage** remains: airport packaging waste is PE/PP-rich, pushing performance toward Scenario C (78% oil yield).
6. **Adopt the V2 model (all char sold commercially)** as the base case for the financial model: all combustible char (~16,088 T/year) is sold at €100/T (conservative fuel pellet pricing), and purchased heating oil supplies all reactor heating. This model treats char as a premium product: fuel-grade quality biochar sold at market prices.
7. **Prioritise char characterisation testing** — bomb calorimetry and elemental analysis on UR-produced char from organic/cellulose feedstock. This is the single most impactful validation for the financial model, as it determines char commercial value and marketability to fuel buyers.

18. References

Urban Rig Test Data

1. **Asada Shokai test programme** (Noda, 2022) — 74-page test report with independent measurement by Land Brain Co., Ltd. Original document: *Asada Shokai data 2.pdf*. Cited for: PE oil yield 77% (test #2), PP oil yield 80% (test #6), PS oil yield 58% (tests #3–4), off-gas CV 5.54 MJ/kg (energy balance tables pp. 35–36), municipal waste tests #14 and #15 (§15).
2. **Urban Rig Recycling Solutions ENG** — OneWorld Corporation product brochure. Cited for: halogen filtration system (pp. 14–15).

Waste Composition Sources

3. **ADEME MODECOM 2017** — France’s national waste characterisation campaign (third edition). Published at ademe.fr; detailed analysis at medias.amf.asso.fr. Cited for: 13-category OMR composition (§1).
4. **MODECOM 2024** — First results at valor3e.fr. Cited for: updated composition trends (§1, note).
5. **PlasticsEurope** (2017) — European plastic packaging waste composition via PMC 8162419. Cited for: polymer mix (LDPE 32%, HDPE 20%, PP 20%, PET 18%, PS 7%, PVC 3%) (§8.1).
6. **EEA** — *Textiles and the Environment* report (2024), eea.europa.eu. Cited for: European textile fibre composition (§4).
7. **Tetra Pak** — Sustainability reports, tetrapak.com. Cited for: composite packaging material breakdown — 75% cardboard, 20% PE, 5% aluminium (§5).
8. **Vertuow** — Sanitary textiles subcategory breakdown, vertuow.com. Cited for: toilet paper 44.3%, paper towels 17.2%, baby diapers 10.7%, wipes 5.4% (§3.1).
9. **ResearchGate / ChemistryExplained.com** — Diaper material composition data. Cited for: cellulose 37%, SAP 31%, PP 16%, LDPE 6% (§3.2).

Pyrolysis Literature

10. **Bridgwater, A.V.** (2012) — “Review of fast pyrolysis of biomass and product upgrading,” *Biomass and Bioenergy*. Cited for: cellulose pyrolysis yields, off-gas cross-validation (§10.10).
11. **Chen, D. et al.** (2014) — “Pyrolysis technologies for municipal solid waste,” *Waste Management*. Cited for: cellulose/putrescible yield ranges (§10.10).
12. **Polymer Degradation and Stability** (2022) — Slow pyrolysis of PET at 400°C, ScienceDirect S0141391022000866. Cited for: PET oil yield 46.7% (§10.4).
13. **PMC 7183261** — PET pyrolysis cross-check. Cited for: PET yield validation (§10.4).
14. **ScienceDirect S0048969723069723** — Two-stage PVC dechlorination pyrolysis. Cited for: PVC oil yield derivation (§10.5).
15. **PMC 12096105** — PVC pyrolysis. Cited for: PVC yield cross-validation (§10.5).

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16. **Fazil, A. et al.** (2021) — *Energies*. Cited for: organic/cellulose char CV estimates (via Annex 3 §3.2.1).
 17. **Ronsse, F. et al.** (2020) — Biochar characterisation. Cited for: organic char CV estimates (via Annex 3 §3.2.1).
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End of Annex 5